

Pinpoint Tests



Pinpoint Test A: Vibration When the Brakes are Applied

Normal Operation

During moderate to heavy braking, noise from the Hydraulic Control Unit (HCU) and pulsation in the brake pedal can be observed. Pedal pulsation coupled with noise during heavy braking or on loose gravel, bumps, wet or snowy surfaces is acceptable and indicates correct functioning of the ABS. Pedal pulsation or steering wheel nibble when the brakes are applied (frequency is proportioned to the vehicle speed) indicates a concern with a brake or suspension component.

PINPOINT TEST A : VIBRATION WHEN THE BRAKES ARE APPLIED

A1 ROAD TEST THE VEHICLE — LIGHT BRAKING

- Road test the vehicle. Warm the brakes by slowing the vehicle from 80 to 32 km/h (50 to 20 mph) using light brake force. At highway speeds of 89-97 km/h (55-60 mph), apply the brake using light pedal force.

Is there a vibration/shudder felt in the steering wheel, seat or brake pedal?

Yes	GO to A4 .
No	GO to A2 .

A2 ROAD TEST THE VEHICLE — MODERATE TO HEAVY BRAKING

- Road test the vehicle. At highway speeds of 89-97 km/h (55-60 mph), apply the brake using a moderate to heavy pedal force.

Is there a vibration/shudder?

Yes	GO to A3 .
No	The concern is not present at this time.

A3 CHECK ABS OPERATION

- NOTE:** During moderate to heavy braking, noise from the HCU and pulsation in the brake pedal can be observed. Pedal pulsation coupled with noise during heavy braking or on loose gravel, bumps, wet or snowy surfaces is acceptable and indicates correct operation of the ABS. Pedal pulsation or steering wheel nibble with the frequency proportional to vehicle speed indicates a concern with a brake or suspension component.

Road test the vehicle and apply the brakes on a dry, firm surface, then apply the brakes on a wet, snowy or loose surface (such as gravel).

Is the vibration/shudder only present on a wet, snowy or loose surface?

Yes	This is a normal operating condition of the ABS.
No	GO to A4 .

A4 CHECK THE FRONT SUSPENSION

- Check the front suspension. Refer to Section 204-00.

Are all the suspension components in satisfactory condition?

Yes	GO to A5 .
No	REPAIR or INSTALL new components as necessary. TEST the system for normal operation.

A5 CHECK THE FRONT BRAKE DISCS

- Inspect the front brake discs. If brake disc thickness is sufficiently above the minimum disc thickness, machine the brake disc. If disc thickness is close to minimum disc thickness, install new brake discs as necessary. Refer to Brake System Inspection in this section.
- Road test the vehicle.

Is the vibration/shudder present?

Yes	GO to A6 .
No	The concern has been repaired.

A6 CHECK THE REAR SUSPENSION

- Check the rear suspension. Only service rear rotors if special issues are present. F150 rear brakes are not sensitive to rotor-induced vibration or shudder. Refer to Section 204-00.

Are all the suspension components in satisfactory condition?

Yes	INSPECT the rear brake discs. If brake disc thickness is sufficiently above the minimum disc thickness, machine the brake disc. If disc thickness is close to minimum disc thickness, install new brake discs as necessary. REFER to Brake System Inspection in this section.
No	REPAIR or INSTALL new components as necessary. TEST the system for normal operation.